

Use the situation below to answer questions 1-3.

A new MacBook computer costs \$950, but depreciates 10.8% each year as new technology comes out.

1. Is this situation exponential growth or decay?
2. Two students, Quinn and Jyquis were debating how to represent a function to model the value, V , after x years. Both of their functions are below. Explain who you agree with and why.

Quinn	Jyquis
$V(x) = 950(1.108)^x$	$V(x) = 1275(0.892)^x$

3. Write the correct function that models the context using both forms.

$V(x) = a(b)^x$	$V(x) = a(1 \pm r)^x$

Use the situation below to answer questions 4-6.

A new school opened with 134 students in 2021 and plans to increase by 26% per year until they reach full capacity.

4. Is this situation exponential growth or decay?

5. Write an equation that models the population of the school, P , after x years since the opening of the school in 2021.
6. Write the correct function that models the context using both forms.

$P(x) = a(b)^x$	$P(x) = a(1 \pm r)^x$

7. A three-bedroom house in the suburbs of Miami sold for \$420,625. If the housing prices are expected to increase 3.75% annually, write a function that models the cost of a house, C , x years after it was sold.
8. A small local company reported a profit of \$89,000 last year, but is struggling due to the pandemic. They are reporting a decrease of 12% each year. Write a function that represents the profit, P , after t years.
9. As of 2020, the world population is 9.23 billion people and is expected to grow at a rate of 0.62% per year. Write an equation to model the population growth, where $P(y)$ is the population in billions of people and y is the time in years.
10. A culture of 15 bacteria grows at a rate of 250% each hour. Tyler wrote the equation $B(t) = 15(2.5)^t$ to represent the population of the bacteria after t hours. Describe and correct the error that Tyler made.